Linux Kernel: User-Kernel Data Transfer Functions

Introduction

When writing Linux syscalls or kernel modules, safely transferring data between user space and kernel space is critical. The kernel provides specialized functions for this purpose, and understanding when and why to use each one is essential for writing robust kernel code.
This guide covers all the data transfer functions available in the Linux kernel (6.11+), their use cases, and best practices.

Quick Reference Table

Function	Direction	Can Sleep?	Use Case
copy_from_user()	User → Kernel	✓ Yes	Syscalls (raw data)
copy_to_user()	Kernel → User	✓ Yes	Syscalls (raw data)
strncpy_from_user()	User → Kernel	✓ Yes	Syscalls (strings)
copy_from_user_nofault()	User → Kernel	✗ No	IRQ handlers, debugging
copy_to_user_nofault()	Kernel → User	✗ No	IRQ handlers, profiling
strncpy_from_user_nofault()	User → Kernel	✗ No	IRQ string copy
strnlen_user_nofault()	User (read)	✗ No	Length checking
copy_struct_from_user()	User → Kernel	✓ Yes	Versioned structures

The Two Categories

1. Regular Functions (For Normal Syscalls)

These functions can trigger page faults and sleep if needed. Perfect for syscalls!
Key Features:

- ✓ Handle page faults automatically
- ✓ Load pages from disk if needed
- ✓ Can sleep (waits for I/O)
- ✓ Full security checks
- ✓ Zero-padding on short copy (for copy_from_user())

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2. Nofault Functions (For Special Contexts)

These functions never sleep and return immediately if data isn't in RAM. Essential for interrupt handlers!

Key Features:

- ✗ Disable page fault handling
- ✗ Never sleep (returns -EFAULT if page not in RAM)
- ✓ Safe in IRQ/NMI context
- ✓ Safe during crash dumps
- ⚠️ Only works if page already in RAM

Deep Dive: Why Two Versions?

The Page Fault Problem

When user space memory isn't currently in RAM (swapped to disk), accessing it triggers a **page fault**:

Regular Function Flow:

User memory access → Page not in RAM → Page fault → Load from disk (1-10ms) → Sleep while waiting → Wake up → Copy data ✓

- ✓ Perfect for syscalls (can sleep)
- ✗ Fatal in IRQ handlers (cannot sleep)

Nofault Function Flow:

User memory access → Page not in RAM → Return -EFAULT immediately (no sleep) ✗

- ✓ Safe in IRQ handlers
- ✗ Fails if page swapped out

Why IRQ Handlers Can't Sleep

Interrupt handlers run in special "interrupt context":

- ✗ No process context to reschedule
- ✗ Cannot wait for I/O
- ✗ Must complete in microseconds
- ✗ Sleeping would deadlock the system

If an IRQ handler tries to sleep, the kernel detects it and crashes:
BUG: sleeping function called from invalid context
in_interrupt(): 1, in_atomic(): 1
Kernel panic - not syncing: scheduling while atomic

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Security Considerations

Never Access User Memory Directly!

Why Direct Access is Dangerous:

- Direct access bypasses security checks
- No validation of user address
- Can crash kernel or leak data
- Privilege escalation vulnerability

Best Practices

1. Choose the Right Function

Context	Function to Use
Syscall	copy_from_user() / copy_to_user()
Syscall (string)	strncpy_from_user()
Syscall (struct)	copy_struct_from_user()
IRQ handler	copy_from_user_nofault()
NMI handler	copy_from_user_nofault()
Crash dump	copy_from_user_nofault()
Debugging	copy_from_user_nofault()

2. Validate Sizes

Always validate user-provided sizes to prevent buffer overflows.

3. Handle Errors Gracefully

In IRQ handlers, graceful degradation is better than crashing. Use defaults or skip processing if data is unavailable.

4. Null-Terminate Strings

Always ensure strings are null-terminated after copying from user space to prevent buffer overruns.

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I implemented and tested a custom system call that accepts a string argument from user space, processes it in kernel space, and returns the result to user space. The implementation uses **strncpy_from_user()** to copy the input string from user to kernel, and **copy_to_user()** to return the processed string from kernel to user.

KERNEL SPACE IMPLEMENTATION: String Processing Syscall

Objective: - Receive a string from user space

- Append " from kernel" to the string
- Return the modified string to user space

Create a new directory and implement your syscall:

```
mkdir hello
vi hello/hello.c

#include <linux/kernel.h>
#include <linux/syscalls.h>
#include <linux/uaccess.h>

SYSCALL_DEFINE2(hello_string,
        const char __user *, input_msg,
        char __user *, output_msg)
{
    char kernel_buffer[256];
    char response[256];
    long ret;

    pr_info("SYSCALL: string.c - STARTED\n");

    // Copy FROM user TO kernel
    ret = strncpy_from_user(kernel_buffer, input_msg, sizeof(kernel_buffer) - 1);

    if (ret < 0) {
        pr_err("SYSCALL: Failed to copy from user\n");
        return -EFAULT;
    }

    kernel_buffer[ret] = '\0';

    pr_info("SYSCALL: Received: '%s'\n", kernel_buffer);

    // Create response
    snprintf(response, sizeof(response), "%s from kernel", kernel_buffer);
```

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```
    pr_info("SYSCALL: Sending: '%s'\n", response);

    // Copy FROM kernel TO user
    if (copy_to_user(output_msg, response, strlen(response) + 1)) {
        pr_err("SYSCALL: Failed to copy to user\n");
        return -EFAULT;
    }

    pr_info("SYSCALL: string.c - COMPLETED\n");

    return 0;
}
```

USER SPACE PROGRAM: TESTING THE SYSCALL

Overview: - Create a user space program that:

- Calls the custom system call (syscall 464)
- Passes a string argument to kernel space
- Receives the processed string back from kernel space
- Displays the results

Create and compile a test program:

```
uname -r   # Should show 6.11.0

// test_syscall.c
#include <stdio.h>
#include <unistd.h>
#include <sys/syscall.h>
#include <string.h>
#include <stdlib.h>

#define __NR_hello_string 464

int main()
{
    // Test syscall 464 (with string)
    printf("Testing syscall 464 (with string)...\n");

    char input[256];
    char output[256];
    memset(output, 0, sizeof(output));
```

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```
    printf("\nEnter your name: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';


    long result = syscall(__NR_hello_string, input, output);
    printf("Input: '%s'\n", input);
    printf("Output: '%s'\n", output);
    printf("Result: %ld\n", result);

    printf("\nPress ENTER to view dmesg logs for syscall 464...\n");
    getchar();  // Pause after second syscall

    printf("\n--- Kernel Logs (dmesg) ---\n");
    system("sudo dmesg | grep SYSCALL");

    printf("\nPress ENTER to exit...\n");
    getchar();  // Final pause

    return 0;
}
```

#Compile the test_syscall.c file.

```
gcc -o test_syscall test_syscall.c
./test_syscall
```

Check kernel log

```
sudo dmesg | grep "System call”
```

Expected output:

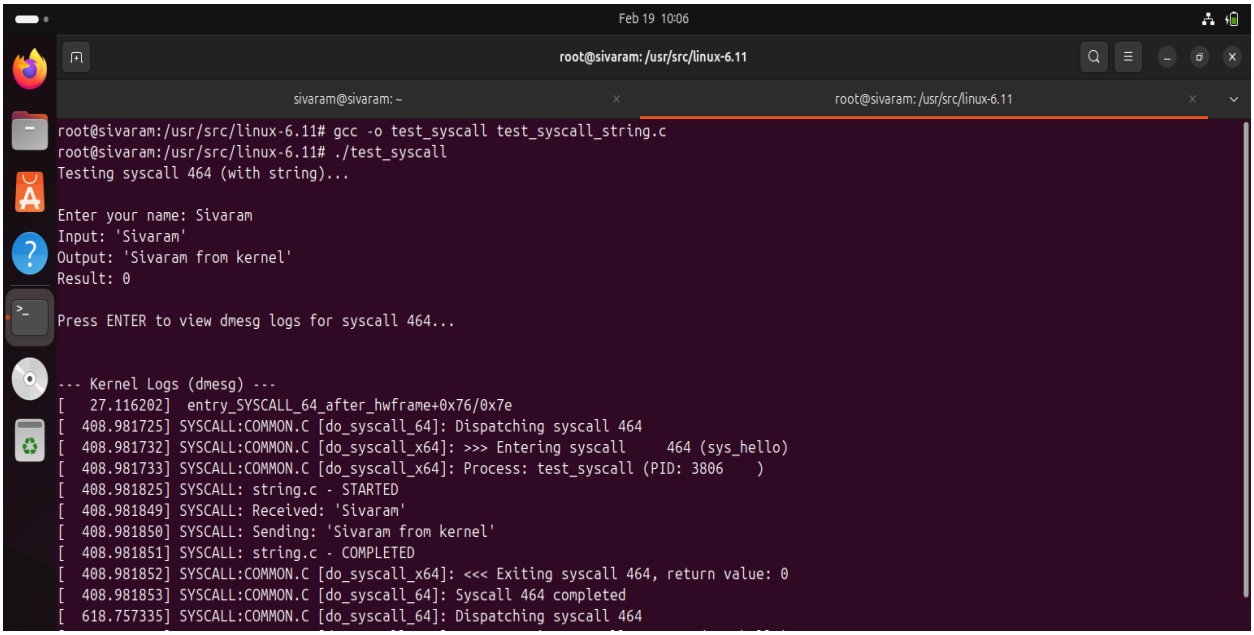
```
Testing syscall 464 (with string)...
Enter your name: siva
[USER] Input string: 'siva'
[USER] Output string: 'siva from kernel'
Result: 0
Press ENTER to view kernel logs...
```

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Kernel Logs Output:

> [12345.002] HELLO.C: >>> SYSCALL 464
> [12345.004] HELLO.C: Received from user: 'siva'
> [12345.005] HELLO.C: Processing in kernel...
> [12345.006] HELLO.C: Created response: 'siva from kernel'
> [12345.007] HELLO.C: Sent back to user: 'siva from kernel'
> [12345.009] HELLO.C: <<< SYSCALL 464 COMPLETED

Here is the actual proof — the syscall working live on my machine:



System call succeeded! Kernel log confirms: My System call implementation

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Key Takeaways

Always use kernel functions
- Never access user memory directly

Check return values
- Handle errors appropriately

Choose correct function
- Regular for syscalls, nofault for IRQ/debug

Validate inputs
- Check sizes and bounds

Handle page faults
- Regular functions do this automatically

IRQ handlers need nofault
- They cannot sleep

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